where ε_{1t} (and later ε_{2t}, ε_{3t}, etc.) is a linear function of ε_t, η_t, C_t^y, and C_t^p and thus uncorrelated with u_t by (3). Using (1) gives

$$y_{St} = \psi\phi^d y_{St} + u_{St} + \varepsilon_{2t}.$$

Hence, using the notation introduced in (13), we have

$$y_{St} = Mu_{St} + \varepsilon_{3t}, \qquad p_t = \mu u_{St} + \varepsilon_{4t}, \qquad y_{\bar{E}t} = (M-1)u_{St} + u_{\bar{E}t} + \varepsilon_{5t}. \tag{59}$$

This gives how idiosyncratic shocks affect y_{St}, p_t, and y_{Et}.

The rest of the proof uses well-known ingredients. We use $\mathbb{E}_T$ for the sample temporal mean, $\mathbb{E}_T[Y_t] := (1/T)\Sigma_{t=1}^T Y_t$. We have $p_t - \psi y_{St} = \varepsilon_t^\psi := C_t^p m^p + \varepsilon_t$, so

$$\psi_T^e - \psi = \frac{\mathbb{E}_T[p_t z_t]}{\mathbb{E}_T[y_{St} z_t]} - \psi = \frac{\mathbb{E}_T[(p_t - \psi y_{St})z_t]}{\mathbb{E}_T[y_{St} z_t]} = \frac{\mathbb{E}_T[\varepsilon_t^\psi u_{\Gamma t}]}{\mathbb{E}_T[y_{St} u_{\Gamma t}]} = \frac{A_T}{D_T}, \tag{60}$$

where $A_T = \mathbb{E}_T[\varepsilon_t^\psi u_{\Gamma t}]$ and $D_T = \mathbb{E}_T[y_{St} u_{\Gamma t}]$. For the denominator, the law of large number gives (as $T \to \infty$)

$$D_T = \mathbb{E}_T[y_{St} u_{\Gamma t}] \xrightarrow{\text{a.s.}} D = \mathbb{E}[y_{St} u_{\Gamma t}].$$

Using (59), we have

$$D = \mathbb{E}[y_{St} u_{\Gamma t}] = \mathbb{E}[(Mu_{St} + \varepsilon_{3t})u_{\Gamma t}] = M\,\mathbb{E}[u_{St} u_{\Gamma t}],$$

which is nonzero because we assumed that $\mathbb{E}[u_{St} u_{\Gamma t}] \neq 0$ and because of the definition of M in (13). For the numerator, the central limit theorem gives the convergence in distribution $\sqrt{T}A_T \xrightarrow{d} \mathcal{N}(0, \sigma_A^2)$, where, using the assumption that ε_t^ψ is homoskedastic conditionally on u_t,

$$\sigma_A^2 = \mathbb{E}\left[\left(\varepsilon_t^\psi\right)^2 u_{\Gamma t}^2\right] = \mathbb{E}\left[\mathbb{E}\left[\left(\varepsilon_t^\psi\right)^2 \mid u_t\right] u_{\Gamma t}^2\right] = \mathbb{E}[\sigma_{\varepsilon^\psi}^2 u_{\Gamma t}^2] = \sigma_{\varepsilon^\psi}^2 \sigma_{u_\Gamma}^2.$$

This implies (by Slutsky's theorem) that $\sqrt{T}(\psi_T^e - \psi) \xrightarrow{d} \mathcal{N}(0, \sigma_\psi^2)$, where $\sigma_\psi = \sigma_A/|D| = (\sigma_{\varepsilon^\psi}\sigma_{u_\Gamma})/(|M\,\mathbb{E}[u_{St} u_{\Gamma t}]|)$. The derivation of $\sigma_{\phi^d}^2$ follows along the same lines. It is detailed in appendix C.

A3. Proof of Proposition 3

We solve for the optimal Γ, which minimizes $\sigma_\psi(\Gamma)$ (and then, automatically, $\sigma_{\phi^d}(\Gamma)$) subject to $\Gamma'\lambda = 0$. Given proposition 2, we want to maximize the squared correlation:

$$\max_\Gamma C(\Gamma) := \operatorname{corr}(u_{St}, u_{\Gamma t})^2 \text{ subject to } \Gamma'\lambda = 0. \tag{61}$$

We next solve this problem. By assumption 3 and (56) we have $\mathbb{E}[u_{St} u_{\Gamma t}] = \sigma_u^2 \Gamma' S$, and hence

$$C(\Gamma) = \frac{\mathbb{E}\left[u_{St} u_{\Gamma t}\right]^2}{\operatorname{var}(u_S)\operatorname{var}(u_{\Gamma t})} = \frac{(\Gamma' S)^2}{(S'S)(\Gamma'\Gamma)}.$$

The problem is invariant to changing Γ into $k\Gamma$ for a nonzero k. So, we can fix $\Gamma' S$ at some value v. Given this, we want the minimum value of $\Gamma'\Gamma$. So, we minimize over Γ the Lagrangian

$$\mathcal{L} = \frac{1}{2}\Gamma'\Gamma - \Gamma'\lambda b - c(\Gamma' S - v), \tag{62}$$

with some Lagrange multipliers b (of dimension $r \times 1$), and c (a scalar). The first-order condition in Γ' is

$$0 = \Gamma - \lambda b - cS \qquad \Longleftrightarrow \qquad \Gamma = cS + \lambda b. \tag{63}$$

We next use the projection operator $Q = I - \lambda(\lambda'\lambda)^{-1}\lambda'$ (see [16]), which satisfies $Q\lambda = 0$. As $\lambda'\Gamma = 0$, we have $Q\Gamma = \Gamma$. So

$$\Gamma = Q\Gamma = cQS + Q\lambda b = cQS. \tag{64}$$

The factor c does not affect the results, as Γ and $c\Gamma$ give the same estimator $\phi_T^{s,e}$, so we may choose $c = 1$ and conclude that $\Gamma = QS$.

A4. *Proof of Lemma 1*

We prove the lemma in the more general heteroskedastic case, with $W = (V^u)^{-1}$. We define, as in (52), $R := (\lambda' W\lambda)^{-1}\lambda' W$ and $Q := I - \lambda R$, dropping the superscripts for concision. We have $Q\lambda = 0$ and $\lambda\check{\eta}_t + \check{u}_t = \lambda(\eta_t + Ru_t) + (I - \lambda R)u_t = \lambda\eta_t + u_t = Y_t$. Finally,

$$\begin{aligned}\mathbb{E}[\check{\eta}_t\check{u}_t'] &= \mathbb{E}[(\eta_t + Ru_t)(Qu_t)'] = \mathbb{E}[Ru_t u_t' Q'] = R\,\mathbb{E}[u_t u_t']Q' = RV^u Q' \\ &= (\lambda' W\lambda)^{-1}\lambda' WV^u Q' = (\lambda' W\lambda)^{-1}\lambda' Q' = (\lambda' W\lambda)^{-1}(Q\lambda)' = 0.\end{aligned}$$

A5. *Proof of Proposition 4*

We use the following regularity assumption.

Assumption 5. The matrices $\mathbb{E}[\check{\eta}_t\check{\eta}_t']$, $\mathbb{E}[(Q^\lambda C_t^y)'(Q^\lambda C_t^y)]$, $\mathbb{E}[C_{St}^{y\prime}C_{St}^{\prime}]$, and $\mathbb{E}[C_t^{p\prime}C_t^p]$ have full rank. The true parameter θ_0 is in the interior of a compact set Θ.

These conditions are mild and technical.[39] We also recall that we maintain the assumption that at the true values, $\psi\phi^d \neq 1$.

We define the function $g_t(\theta) = (g_t^k(\theta))_{k=1,2,3}$ as

$$g_t^1(\theta) := \left(-\check{y}_t + \check{\lambda}\check{\eta}_t(m^y) + \check{C}_t^y m^y\right)'\check{C}_t^y, \tag{65}$$

$$g_t^2(\theta) := \left(-p_t + \psi y_{St} + b^p\check{\eta}_t(m^y) + m^{p\prime}C_t^{p\prime}\right)\left(z_t(m^y), \check{\eta}_t(m^y)', C_t^p\right), \tag{66}$$

$$g_t^3(\theta) := \left(-y_{Et} + \phi^d p_t + b^y\check{\eta}_t(m^y) + C_{Et}^y m^y\right)(z_t(m^y), \check{\eta}_t(m^y)'). \tag{67}$$

We again use $\mathbb{E}_T$ for the sample temporal mean, $\mathbb{E}_T[Y_t] := (1/T)\Sigma_{t=1}^T Y_t$, so that, given a sample of size T, our estimator θ_T^e solves $\mathbb{E}_T[g_t(\theta_T^e)] = 0$. Among other things, the proof verifies that, for large T, the solution exists and is unique.

At the true value, $\mathbb{E}[g_t(\theta_0)] = 0$.—Indeed, $g_t^1(\theta_0) = -\check{u}_t'\check{C}_t^y$, as spelled out in equation (73). This implies that $\mathbb{E}[g_t^1(\theta_0)] = 0$, by (3) and the fact that $\check{u}_t = Q^\lambda u_t$.

[39] For instance, if $\mathbb{E}[\check{\eta}_t\check{\eta}_t']$ does not have full rank $r - 1$, one can simply use a lower-dimensional factor model and achieve full rank.

Likewise, $g_t^2(\theta_0) = -\varepsilon_t^{\perp}(z_t, \check{\eta}_t', C_t^{p\prime})$, so $\mathbb{E}[g_t^2(\theta_0)] = 0$, by the maintained assumptions of section II.A, which imply that $\varepsilon_t, \eta_t \perp C_t^{p\prime}, u_t, \check{u}_t, z_t$ as $z_t = \Gamma' u_t$; lemma 1, which implies that $\check{\eta}_t := \eta_t + R^{\check{\lambda}} u_t \perp C_t^{p\prime}, \check{u}_t, z_t$ as additionally $z_t = \Gamma' \check{u}_t$; and the fact that $\varepsilon_t^{\perp} = \varepsilon_t - b^e \check{\eta}_t \perp \check{\eta}_t$, which implies that $\varepsilon_t^{\perp} \perp C_t^{p\prime}, z_t, \check{\eta}_t$. Similarly, $\mathbb{E}[g_t^3(\theta_0)] = 0$.

The solution of $\mathbb{E}[g_t(\theta)] = 0$ *is unique.*—The high-level idea is that the equation $\mathbb{E}[g_t^1(\theta)] = 0$ is a linear equation in m^y and identifies m^y given our full-rank assumption 5. Then, given m^y, the other equations ($\mathbb{E}[g_t^k(\theta)] = 0$ for $k = 2, 3$) are linear in all the other parameters in θ, so that unicity of the solution is guaranteed as those linear equations have full rank. We spell out the details in appendix C.

The estimator is consistent.—Our estimator θ_T^e is the solution of $\mathbb{E}_T[g_t(\theta)] = 0$.[40] Note that θ_T^e is equivalent to the GMM estimator that minimizes $\mathbb{E}_T[g_t(\theta)]'\mathbb{E}_T[g_t(\theta)]$, as indeed it achieves $\mathbb{E}_T[g_t(\theta)] = 0$. We thus proceed to apply general GMM results (Hansen 1982; Newey and McFadden 1994). In particular under the regularity conditions and i.i.d. sampling, by theorem 2.6 in Newey and McFadden (1994), the estimator is consistent.

The estimator is $\sqrt{T}$*-asymptotically normal.*—Under the regularity conditions we assumed (and in particular that $\psi\phi^d \neq 1$, which ensures bounded gradients), by theorem 3.4 in Newey and McFadden (1994), $\sqrt{T}(\theta_T^e - \theta)$ converges in distribution to a normal distribution with mean zero and positive variance covariance matrix V^θ.

Derivation of the asymptotic variance of ψ_T^e.—It turns out that it is enough to zoom in on the first component of (66)—that is, the moment $\mathbb{E}_T[g_t^{2,z}(\theta)] = 0$, where

$$g_t^{2,z} = \left(-p_t + \psi y_{St} + b^p \check{\eta}_t(m^y) + C_t^p m^p\right) z_t(m^y). \tag{68}$$

As part of our application of Newey and McFadden's (1994) theorem 6.1, we now show that $\mathbb{E}[g_{\theta_k}^{2,z}]$ has only one nonzero component k, the one corresponding to $\theta_k = \psi$.

We take all derivatives at the true value, θ_0. For notational simplicity, we drop the t when the meaning is clear; for example, we write $g_\psi^{2,z}$ for $(d/d\psi)g_t^{2,z}$. As $g_\psi^{2,z} = y_{St} z_t$, it holds that

$$\mathbb{E}\left[g_\psi^{2,z}\right] = \mathbb{E}[y_{St} z_t] = M\,\mathbb{E}[u_{St} z_t]. \tag{69}$$

Next, $\check{\eta}_t(m^y) := R^{\check{\lambda}}(y_t - C_t^y m^y)$ and $z_t(m^y) := \Gamma'(y_t - C_t^y m^y)$ yield

$$\frac{d}{dm^y}\check{\eta}_t(m^y) = -R^{\check{\lambda}} C_t^y, \qquad \frac{d}{dm^y} z_t(m^y) = -\Gamma' C_t^y, \tag{70}$$

and thus

$$\mathbb{E}\left[\left(\frac{d}{dm^y}\check{\eta}_t(m^y)\right) z_t\right] = 0.$$

This allows us to verify that all the other derivatives of $\mathbb{E}[g_{\theta_k}^{2,z}]$ for the other components k of θ (except ψ) are zero. For instance, $(d/db^p)g_t^{2,z} = \check{\eta}_t z_t$, so $\mathbb{E}[g_{b^p}^{2,z}] = \mathbb{E}[\check{\eta}_t z_t] = 0$. Likewise, $\mathbb{E}[g_{\phi^d}^{2,z}] = 0$ and $\mathbb{E}[g_{b^y}^{2,z}] = 0$. More involved is the following:

[40] Moreover, the solution exists and is unique, by the same reasoning as for $\mathbb{E}[g_t(\theta)] = 0$, replacing $\mathbb{E}$ by $\mathbb{E}_T$ when needed.

$$\begin{aligned}\mathbb{E}\left[g^{2,z}_{m_y}\right] &= \mathbb{E}\left[\left(\frac{d}{dm_y}\left(-p_t + \psi y_{St} + b^p \check{\eta}_t(m^y) + C^p_t m^p\right)\right) z_t\right.\\ &\qquad \left. + \left(-p_t + \psi y_{St} + b^p \check{\eta}_t(m^y) + C^p_t m^p\right)\left(\frac{d}{dm^y} z_t\right)\right]\\ &= b^p\, \mathbb{E}\left[\left(\frac{d}{dm^y}\check{\eta}_t(m^y)\right) z_t\right] + \mathbb{E}[(-\varepsilon_t)(-\Gamma' C^y_t)] = 0 + 0 = 0.\end{aligned}$$

Hence, we just showed that $\mathbb{E}[g^{2,z}_\theta]$ has only one nonzero component, the $\mathbb{E}[g^{2,z}_{\theta_k}]$ corresponding to $\theta_k = \psi$. Then, applying Newey and McFadden's (1994) theorem 6.1 allows us to conclude that[41]

$$\sigma^2_\psi = \frac{\mathrm{var}(g^{2,z}_t)}{\left(\mathbb{E}\left[g^{2,z}_\psi\right]\right)^2}. \tag{71}$$

Given that $g^{2,z}_t = -\varepsilon^\perp_t z_t$, and using the assumption that $\varepsilon^\perp_t$ is homoskedastic conditional on u_t,

$$\mathrm{var}(g^{2,z}_t) = \mathbb{E}\left[\left(\varepsilon^\perp_t z_t\right)^2\right] = \mathbb{E}\left[\mathbb{E}\left[\left(\varepsilon^\perp_t\right)^2 | u_t\right] z^2_t\right] = \mathbb{E}\left[\left(\varepsilon^\perp_t\right)^2\right] \mathbb{E}[z^2_t],$$

so that, using (69),

$$\sigma^2_\psi = \frac{\mathrm{var}(\varepsilon^\perp_t)\, \mathbb{E}[z^2_t]}{\left(M\, \mathbb{E}[u_{St} z_t]\right)^2}.$$

This is the asymptotic variance σ^2_ψ that a "naive" IV estimator would report, ignoring that m^y and $\check{\eta}_t$ had to be estimated. If we did not have to estimate m^y and $\check{\eta}_t$, we would obtain the same asymptotic error σ^2_ψ.

The optimal Γ *remains as in proposition 3.*—This comes straightforwardly from the fact that the variance of the estimator is proportional to $\mathbb{E}[u^2_{\Gamma t}] / \mathbb{E}\left[u_{St} u_{\Gamma t}\right]^2$, which is minimized at Γ^*.

Calculating the error on $\phi^{d,e}_T$.—The argument is exactly the same as for ψ. It is spelled out in appendix C.

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[41] In the notations of Newey and McFadden's (1994) theorem 6.1, we have $G_\gamma = 0$—i.e., the gradient of the moment condition $g^{2,z}$ with respect to the nuisance parameters $\gamma := \theta \backslash \{\psi\}$ is zero.